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UNIVERSITY OF NAIROBI

BSC. Computer Science Year 1. SMA3211 Linear Algebra 2025/2026

Continuous Assessment Test (C.A.T)

Date: April 24, 2026, Time: 12:00 PM - 13:00 PM

Answer all the questions in the spaces provided.

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1. The system of equations

$$x + y + z = 6$$
$$x + 4y + 6z = 20$$
$$x + 4y + \lambda z = \mu$$

has NO solution for values of λ and μ given by;

(a) $\lambda = 6, \mu = 20$ (b) $\lambda \neq 6, \mu = 20$ (c) $\lambda = 6, \mu \neq 20$ (d) $\lambda \neq 6, \mu \neq 20$

Justify! (5 MARKS)

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$x+y+z=6$
$x+4y+6z=20$
$x+4y+\lambda z=\mu$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 4 & 6 & 20 \\ 1 & 4 & \lambda & \mu \end{array}\right]$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 3 & 5 & 14 \\ 0 & 3 & \lambda-1 & \mu-6 \end{array}\right]$$

$R_3 \to R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 3 & 5 & 14 \\ 0 & 0 & \lambda-6 & \mu-20 \end{array}\right]$$

$\lambda - 6 = \mu - 20$

$x+4y+6z=20$

$x+4y+(\lambda-6)=\mu-20$

$6z = \lambda - 6$

$\lambda = 6$

$\mu - 20 = 20$

~~The system~~ is inconsistent hence has no solution ~~from the second~~

2. The system of linear equations

$$\begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 14 \end{pmatrix}$$

has:

(a) A unique solution (b) Infinitely many solution (c) No solution (d) Exactly two solutions.

And state whether the system is consistent or inconsistent.

Reason: (5 MARKS)

$\begin{bmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix}$

$R_1 \leftrightarrow R_3$

$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$

$R_2 \to R_2 - 3R_1$ $3-3(1)=0$

$R_3 \to R_3 - 2R_1$ $1-2(1)=-1$

$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -6 & -14 \\ 0 & -3 & -7 \end{bmatrix}$

$R_3 \to R_3 - \frac{1}{2}R_2$

$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -6 & -14 \\ 0 & 0 & 3 \end{bmatrix}$ $-3-(-\frac{1}{2}(6))=0$ $-7+7=0$

$\begin{bmatrix} 1 & 2 & 5 & 14 \\ 3 & 0 & 1 & -4 \\ 2 & 1 & 3 & 5 \end{bmatrix}$ $-4-3(14)=-46$

$5\begin{bmatrix} 1 & 2 & 5 & 14 \\ 3 & 0 & 1 & -4 \\ 2 & 1 & 3 & 5 \end{bmatrix}$

-4

14

$\begin{bmatrix} 1 & 2 & 5 & 14 \\ 0 & -6 & -14 & -46 \\ 0 & -3 & -7 & -23 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & 5 & 14 \\ 0 & -6 & -14 & -46 \\ 0 & 0 & 3 & 0 \end{bmatrix}$

$3z = 0$

$z = 0$

$-6y = 46$

$y = 7\frac{2}{3}$

$x + 2(7\frac{2}{3}) + 0 = 14$

$x = 1\frac{1}{3}$

The system is consistent

3. For matrices of the same dimension M, N and scalars c, which of these properties DOES NOT ALWAYS hold!

(a) $(M^T)^T = M$ (b) $(cM)^T = c(M)^T$ (c) $(M+N)^T = M^T + N^T$ (d) $MN = NM$

Clearly Show!(5 MARKS):

Let M be $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Let N be $\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$

M

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \neq \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$(1 \times 4) + (2 \times 3)$

$= 10$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 4 \\ 24 & 10 \end{bmatrix}$

$\begin{bmatrix} 10 & 24 \\ 4 & 10 \end{bmatrix}$

$\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 16 & 10 \end{bmatrix}$

$\begin{bmatrix} 10 & 24 \\ 4 & 10 \end{bmatrix} \neq \begin{bmatrix} 10 & 6 \\ 16 & 10 \end{bmatrix}$

hence $MN \neq NM$ hence does not hold

4. For $A = \begin{bmatrix} 1 & tanx \\ -tanx & 1 \end{bmatrix}$, the determinant of $A^T A^{-1}$ is:

(a) sec^2x (b) $cos4x$ (c) 1 (d) 0

How! (5 MARKS)

The determinant of a matrix multiplied by its inverse is always 1.

$\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ ✓

$A \cdot A^{-1} = I$

$\det(A \cdot A^{-1}) = \det(I)$

$\det A = |A| \quad A^{-1} = \frac{1}{|A|}$

$\det A \cdot \det A^{-1} = \det I$

$\det A \cdot \det A^{-1} = 1$ 1

$\det A = \frac{1}{A^{-1}}$

$1 = \det A \cdot \det A^{-1}$

$adj = \begin{bmatrix} -36 & 0 & -10 \\ -6 & -56 & -4 \\ -16 & -10 & 25 \end{bmatrix}$

$A^{-1} = \frac{1}{-58}\begin{bmatrix} -36 & 0 & 10 \\ -6 & -56 & -4 \\ -16 & -10 & 25 \end{bmatrix}$

5. Find A^{-1} for

$A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$

(5 MARKS)

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$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -4 & 2 \\ 2 & 3 & -4 \end{bmatrix}$

Reduced Echlon Form

$R_3 \to R_3 \leftarrow\!\to R_1$

$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -4 & 2 \\ 2 & 3 & -4 \end{bmatrix}$ $5 - 2(-1)$ $5 + 2 =$ $3 - 2(-4)$ $3 + 8$

$R_3 \to R_3 - 2R_1$

$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -4 & 2 \\ 0 & 5 & -14 \end{bmatrix}$ $3 + 2$ $3 - 2(5)$ $-4 - 10$

$R_2 \to R_3$

$R_2 \to 1/4\, R_2$

$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -1 & 1/2 \\ 0 & 5 & -14 \end{bmatrix}$ $-14 - 2\frac{1}{2}$ $-11\frac{1}{2}$

$R_3 \to R_3 - 5(R_2)$

$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -1 & 1/2 \\ 0 & 0 & -11\frac{1}{2} \end{bmatrix}$

$1 \times -1 \times -11\frac{1}{2}$

$\det = -58$

$\det(A) = -58 \quad A^{-1} = \frac{1}{\det A}(adj)$

$\begin{bmatrix} + & - & + \\ 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{bmatrix} = 2 \cdot$

$= 2 \cdot \begin{bmatrix} -4 & 2 \\ -1 & 5 \end{bmatrix} - 3\begin{bmatrix} 0 & 2 \\ 1 & 5 \end{bmatrix} + -4\begin{bmatrix} 0 & -4 \\ 0 & -1 \end{bmatrix}$

$= 2(-18) - 3(0-2) + -4(0-(-4)$

$= -58$

$\det(A) = -58$

$A^{-1} = \frac{1}{\det A}[adj]$

$adj =$

$2 \cdot \begin{bmatrix} -4 & 2 \\ -1 & 5 \end{bmatrix} - 3\begin{bmatrix} 0 & 2 \\ 1 & 5 \end{bmatrix} + -4\begin{bmatrix} 0 & -4 \\ 0 & -1 \end{bmatrix}$

$0 \quad +\,-4\begin{bmatrix} 2 & -4 \\ 1 & 5 \end{bmatrix} \quad -2\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$

$+1\begin{bmatrix} 3 & -4 \\ -4 & 2 \end{bmatrix} \quad -1\begin{bmatrix} 2 & -4 \\ 0 & 2 \end{bmatrix} \quad +5\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$

6. The Euclidean norm (length of the vector) $(4, -2, -6)^T$ is; (4 MARKS)

(a) $\sqrt{48}$ (b) $\sqrt{24}$ (c) $\sqrt{12}$ (d) $\sqrt{56}$

$length = \sqrt{4^2 + (-2)^2 + (-6)^2}$

$length = \sqrt{56}$

$2 \cdot \begin{bmatrix} -4 & 2 \\ -1 & 5 \end{bmatrix} - 3\begin{bmatrix} 0 & 2 \\ 1 & 5 \end{bmatrix} + -4\begin{bmatrix} 0 & -4 \\ 1 & -1 \end{bmatrix}$

$2(-18) - 3(0-2) + -4(0-(-4)$

$-36 - 6 - 16 = -58$

$= -58 = \det$

$M = \sqrt{4^2 + (-2)^2 + (-6^2)}$

$M = \sqrt{56}$

7. Let X be a square matrix. Consider the following two statements on X.

   I. X is invertible.

   II. Determinant of X is non-zero.

   Which one of the following is true;

   (a) $I \implies II : II \nRightarrow I$ (b) $II \implies I : I \nRightarrow II$ (c) $I \nRightarrow II : II \nRightarrow I$ (d) I and II are equivalent statements.

   Explain! (6 MARKS)

A matrix (square) X is invertible only if the determinant is ~~zero~~ $\neq$ zero and the determinant of the same matrix is $\neq 0$ and thus can be inverted.

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\det = (1 \times 4) - (3 \times 2)$

$4 - 6$

$= -2$

$\frac{1}{-2}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$= \begin{bmatrix} -\frac{1}{2} & -1 \\ -3\frac{1}{2} & -2 \end{bmatrix}$

but if the ~~square m~~ determinant of the square matrix is 0

$\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix} \Rightarrow \det(A) = (2 \cdot 0) - (1 \cdot 0)$

$= 0$

hence

$\frac{1}{0}\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}$ hence an inverse cannot be found.

$X \cdot X^{-1} = I$

$\det(X \cdot X^{-1}) = \det(I)$

$\det(X) \cdot \det(X^{-1}) = 1$

8. The matrix P is the inverse of a matrix Q. If I denotes the identity matrix, which of the following options is correct;

   (a) $PQ = I, QP \neq I$ (b) $QP = I, PQ \neq I$ (c) $PQ = I, QP = I$ (d) $PQ = QP = I$

   Show! (5 MARKS)

$X^{-1} = \frac{1}{\det(X)}$ .dad;

$A \cdot A^{-1} = I$ $Q \cdot P = I$ We know that $A \cdot A^{-1} = I$ P is A^{-1} Q is A	We also know that For Matrices of the same dimension $MN \neq NM$ The result of the transformation of matrix $(M \cdot N)$ is not the same as $(N \cdot M)$ Hence $A^{-1} \cdot A \neq I$ $\therefore PQ \neq I$ but $QP = I$	Hence $QP = I$, $PQ \neq I$

NOTE!

(i.) "Remember to stay in your own row and column during the exam - you are the determinant of your success!"

(ii.) "Don't overthink - this is not Gauss's birthday party. Just row-reduce your stress and focus!"

END

Question 5

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{bmatrix}$$

$R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -4 & 2 \\ 2 & 3 & -4 \end{bmatrix}$$

$R_3 \to R_3 - 2R_1$

$$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -4 & 2 \\ 0 & 5 & -14 \end{bmatrix}$$

~~$R_3 \to R_3 - 1/$~~

$R_2 \to \frac{1}{4} R_2$

$$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -1 & 1/2 \\ 0 & 5 & -14 \end{bmatrix}$$

$R_3 \to R_3 - 5R_2$

$$\begin{bmatrix} 1 & -1 & 5 \\ 0 & -1 & 1/2 \\ 0 & 0 & -11\,1/2 \end{bmatrix}$$

$\det = -11\,1/2$

$$1\begin{bmatrix} -1 & 1/2 \\ 0 & -11\,1/2 \end{bmatrix} = 11\,1/2$$

$$-1\begin{bmatrix} 0 & 1/2 \\ 0 & -11\,1/2 \end{bmatrix} = 0$$

$$+5\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} = 0$$

-0

$$-1\begin{bmatrix} 1 & 1/2 \\ 0 & -11\,1/2 \end{bmatrix} \quad -11\,1/2 \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$$

$$1/2 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$adj = \begin{bmatrix} 11\,1/2 & 0 & 0 \\ 0 & -16\,1/2 & 0 \\ 0 & 0 & 11\,1/2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-11\,1/2} \begin{bmatrix} 11\,1/2 & 0 & 0 \\ 0 & -16\,1/2 & 0 \\ 0 & 0 & 11\,1/2 \end{bmatrix}$$